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西交利物浦大學

INT305 Machine Learning

Lecture 4

Support Vector Machine, SVM Loss and Softmax Loss

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Binary Classification with a Linear Model

- Classification: Predict a discrete-valued target
- Binary classification: Targets $t \in \{-1, +1\}$
- Linear model:

$$z = \mathbf{w}^\top \mathbf{x} + b$$

$$y = \text{sign}(z)$$

- Question: How should we choose $\mathbf{w}$ and b ?

Zero-One Loss

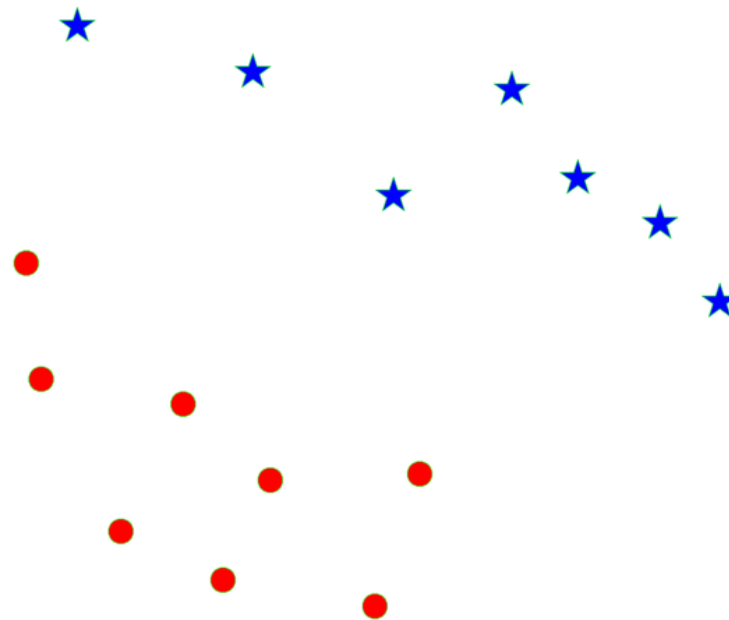
- We can use the 0-1 loss function, and find the weights that minimize it over data points

$$\begin{aligned}\mathcal{L}_{0-1}(y, t) &= \begin{cases} 0 & \text{if } y = t \\ 1 & \text{if } y \neq t \end{cases} \\ &= \mathbb{I}\{y \neq t\}.\end{aligned}$$

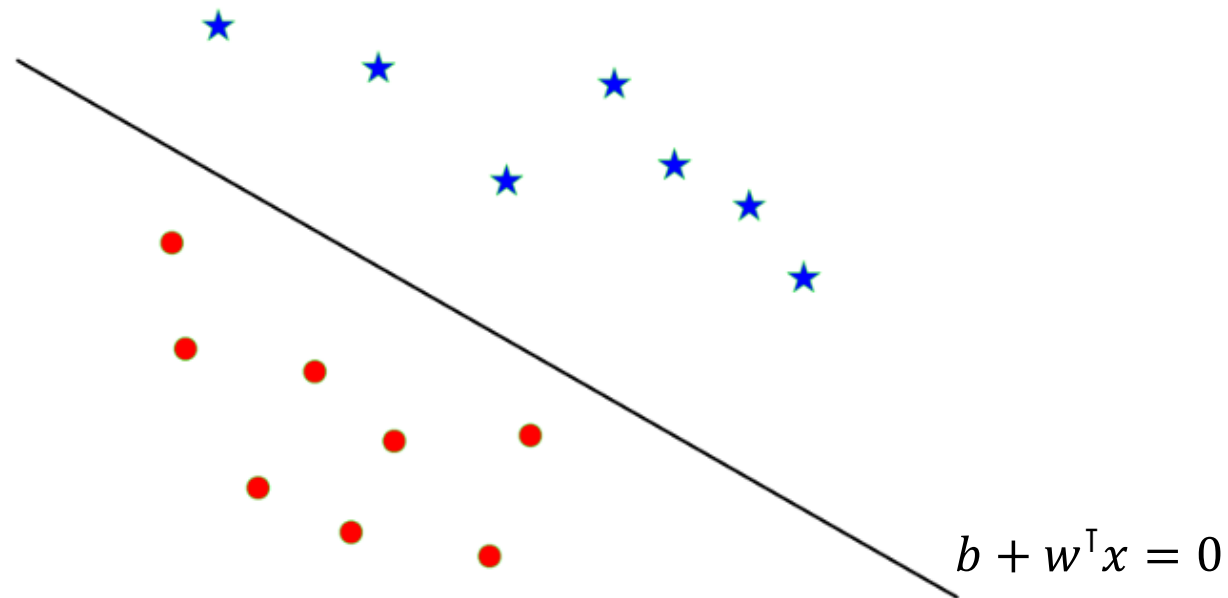
- But minimizing this loss is computationally difficult, and it can't distinguish different hypotheses that achieve the same accuracy.
- We investigated some other loss functions that are easier to minimize, e.g., logistic regression with the cross-entropy loss $\mathcal{L}_{\text{CE}}$.
- Let's consider a different approach, starting from geometry of binary classifiers.

Separating Hyperplanes

Suppose we are given these data points from two different classes and want to find a linear classifier that separates them.

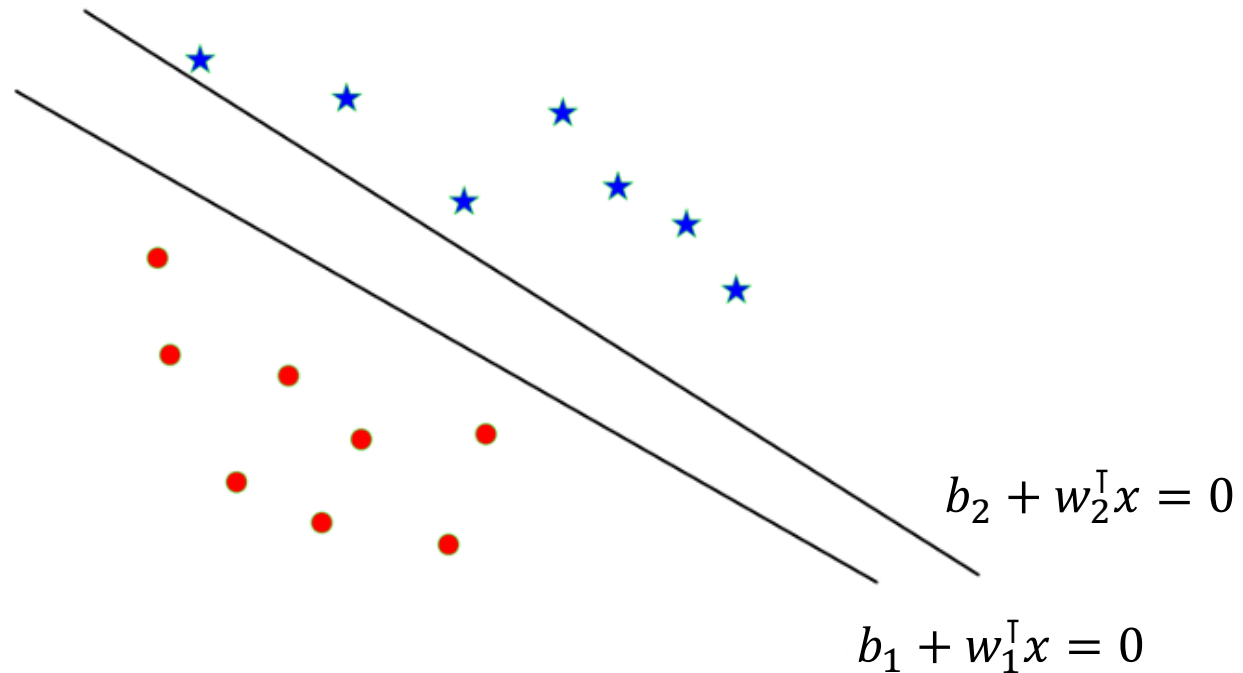


Separating Hyperplanes



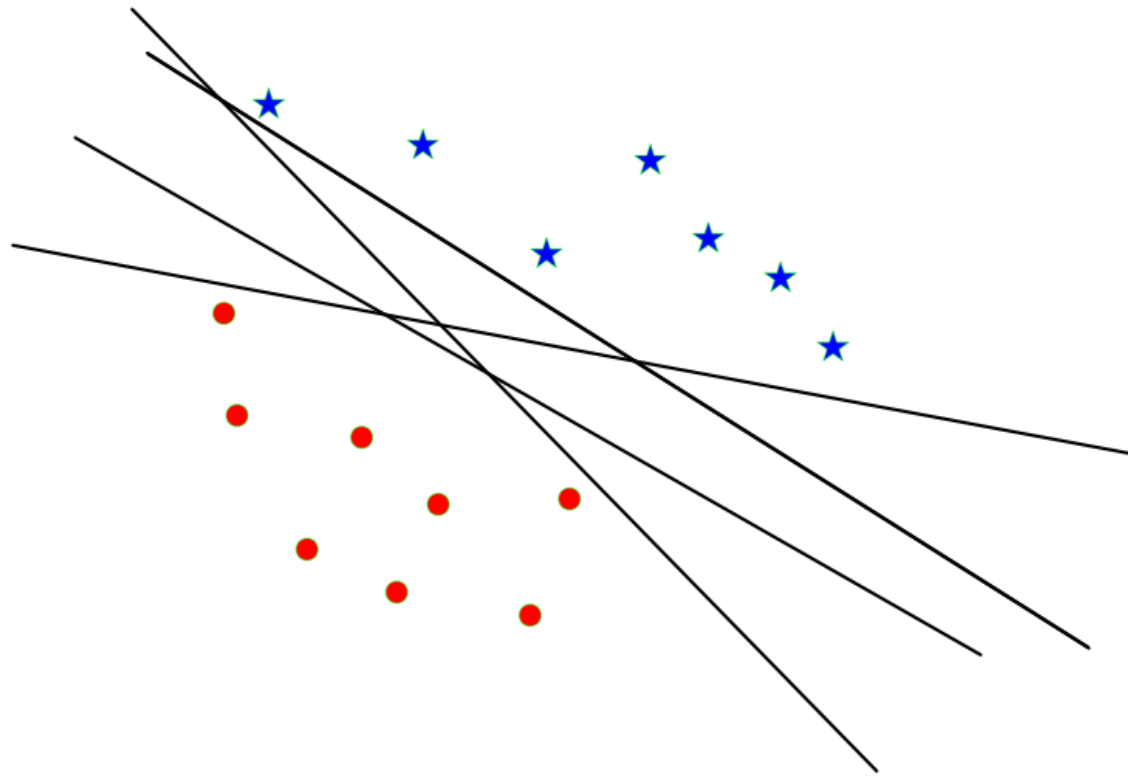
- The decision boundary looks like a line because $\mathbf{x} \in \mathbb{R}^2$, but think about it as a $D - 1$ dimensional hyperplane.
- Recall that a hyperplane is described by points $\mathbf{x} \in \mathbb{R}^D$ such that $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b = 0$.

Separating Hyperplanes



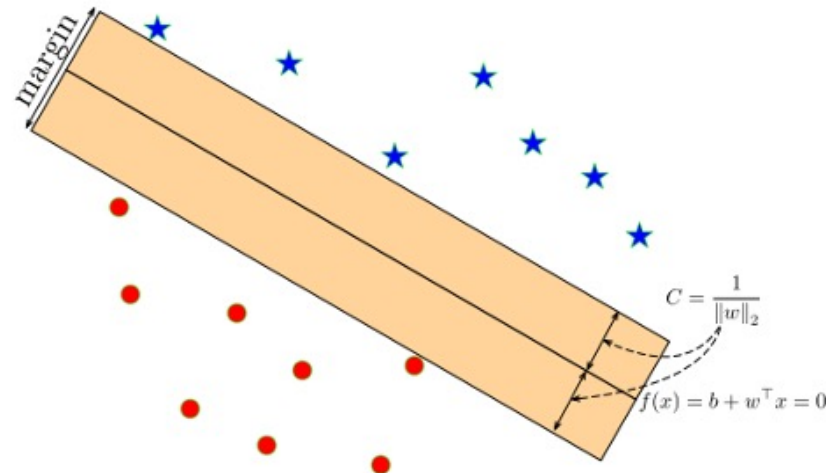
- There are multiple separating hyperplanes, described by different parameters $(\mathbf{w}, b)$.

Separating Hyperplanes



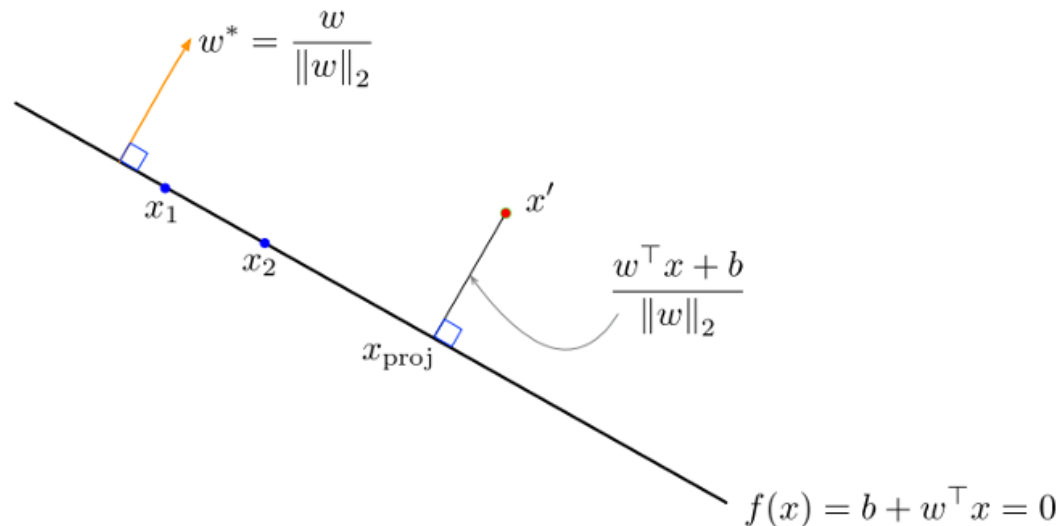
Optimal Separating Hyperplane

Optimal Separating Hyperplane: A hyperplane that separates two classes and maximizes the distance to the closest point from either class, i.e., maximize the **margin** of the classifier.



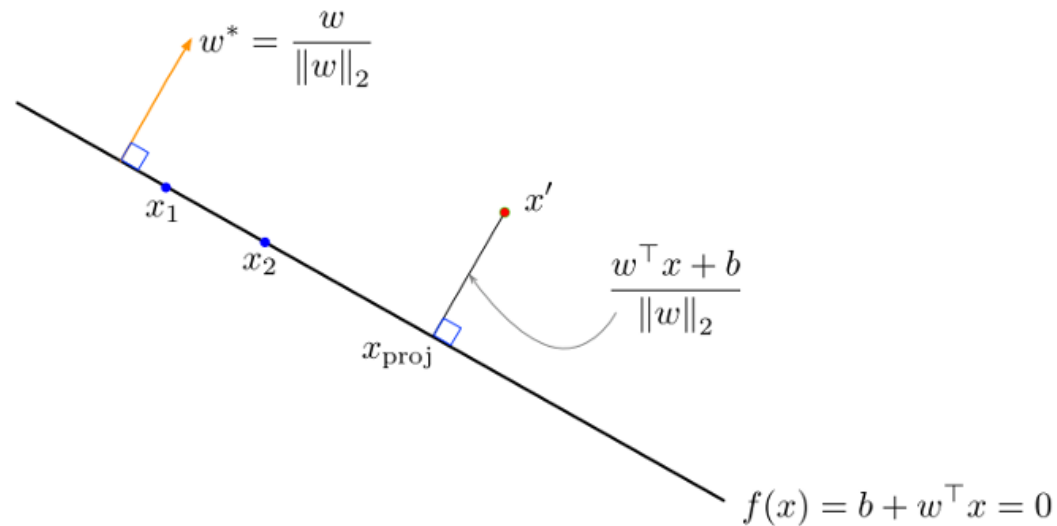
Intuitively, ensuring that a classifier is not too close to any data points leads to better generalization on the test data.

Geometry of Points and Planes



- Recall that the decision hyperplane is orthogonal (perpendicular) to $\mathbf{w}$.
- The vector $\mathbf{w}^* = \frac{\mathbf{w}}{\|\mathbf{w}\|_2}$ is a unit vector pointing in the same direction as $\mathbf{w}$.
- The same hyperplane could equivalently be defined in terms of $\mathbf{w}^*$.

Geometry of Points and Planes



The (signed) distance of a point $\mathbf{x}'$ to the hyperplane is

$$\frac{\mathbf{w}^T \mathbf{x}' + b}{\|\mathbf{w}\|_2}$$

Maximizing Margin as an Optimization Problem

- Recall: the classification for the i -th data point is correct when

$$\text{sign}(\mathbf{w}^\top \mathbf{x}^{(i)} + b) = t^{(i)}$$

- This can be rewritten as

$$t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b) > 0$$

- Enforcing a margin of C :

硬间隔

$$t^{(i)} \cdot \underbrace{\frac{(\mathbf{w}^\top \mathbf{x}^{(i)} + b)}{\|\mathbf{w}\|_2}}_{\text{signed distance}} \geq C$$

Maximizing Margin as an Optimization Problem

Max-margin objective:

$$\begin{aligned} & \max_{\mathbf{w}, b} C \\ & \text{s. t. } \frac{t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b)}{\|\mathbf{w}\|_2} \geq C \quad i = 1, \dots, N \end{aligned}$$

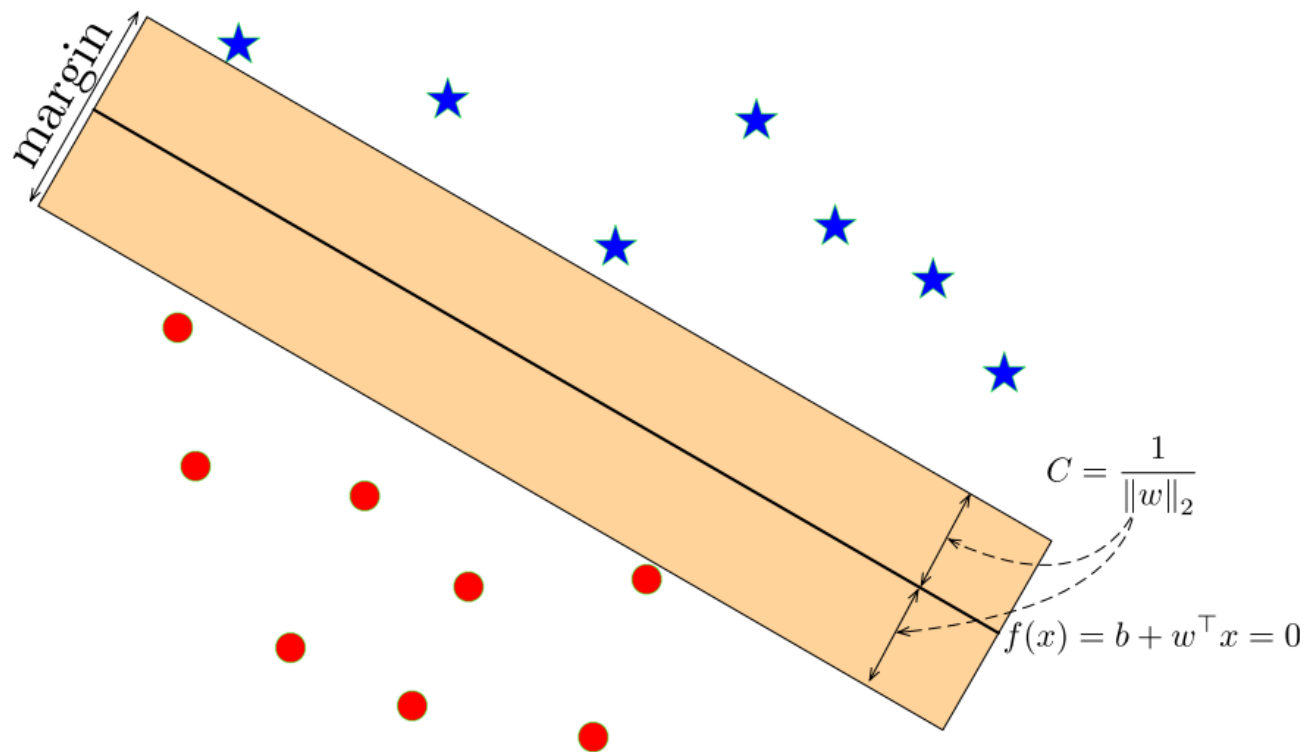
Plug in $C = 1/\|\mathbf{w}\|_2$ and simplify:

$$\underbrace{\frac{t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b)}{\|\mathbf{w}\|_2} \geq \frac{1}{\|\mathbf{w}\|_2}}_{\text{geometric margin constraint}} \iff \underbrace{t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b) \geq 1}_{\text{algebraic margin constraint}}$$

Equivalent optimization objective:

$$\begin{aligned} & \min \|\mathbf{w}\|_2^2 \\ & \text{s. t. } t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b) \geq 1 \quad i = 1, \dots, N \end{aligned}$$

Maximizing Margin as an Optimization Problem



Maximizing Margin as an Optimization Problem

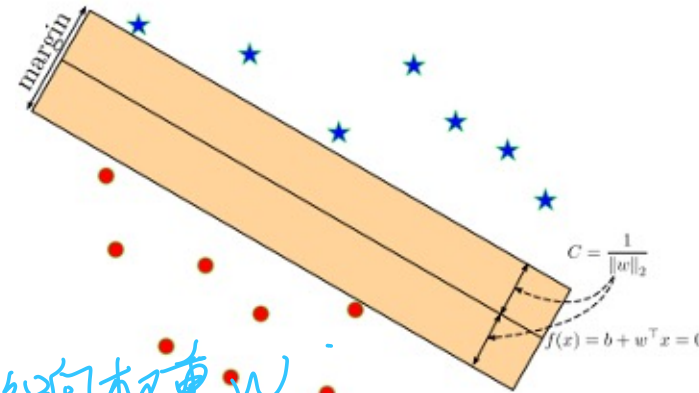
Algebraic max-margin objective:

$$\min \|\mathbf{w}\|_2^2$$

$$\text{s. t. } t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b) \geq 1 \quad i = 1, \dots, N$$

支持向量

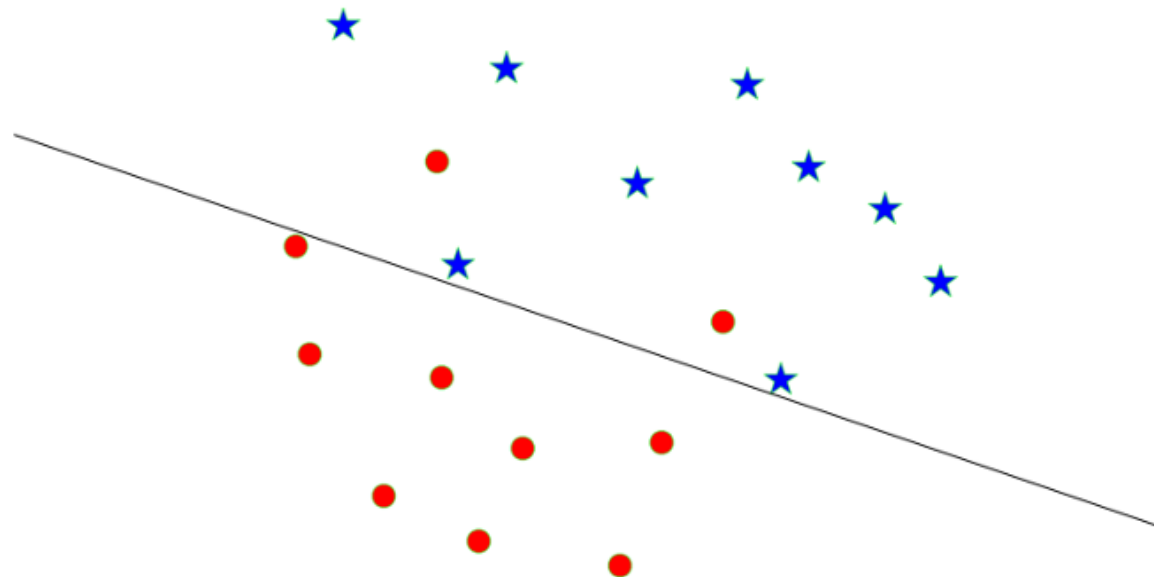
→ 只有边际约束为1的样本影响权重 $\mathbf{w}$



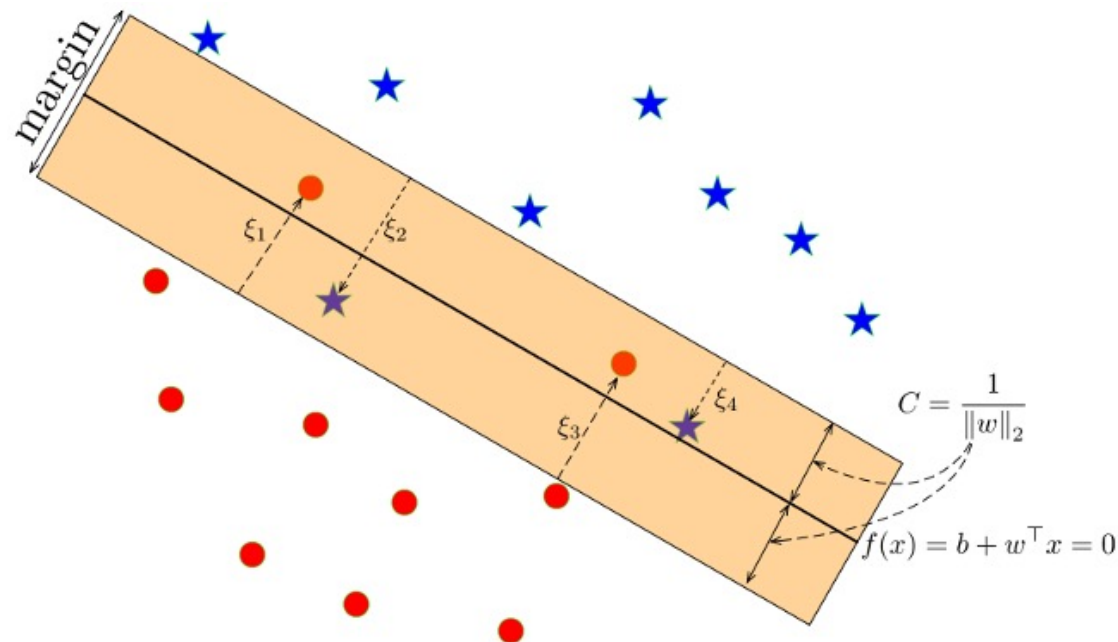
- Observe: if the margin constraint is not tight for $\mathbf{x}^{(i)}$, we could remove it from the training set and the optimal $\mathbf{w}$ would be the same.
- The important training examples are the ones with algebraic margin 1, and are called **support vectors**.
- Hence, this algorithm is called the (hard) **Support Vector Machine (SVM)** (or Support Vector Classifier).
- SVM-like algorithms are often called **max-margin** or **large-margin**.

Non-Separable Data Points

How can we apply the max-margin principle if the data are **not** linearly separable?



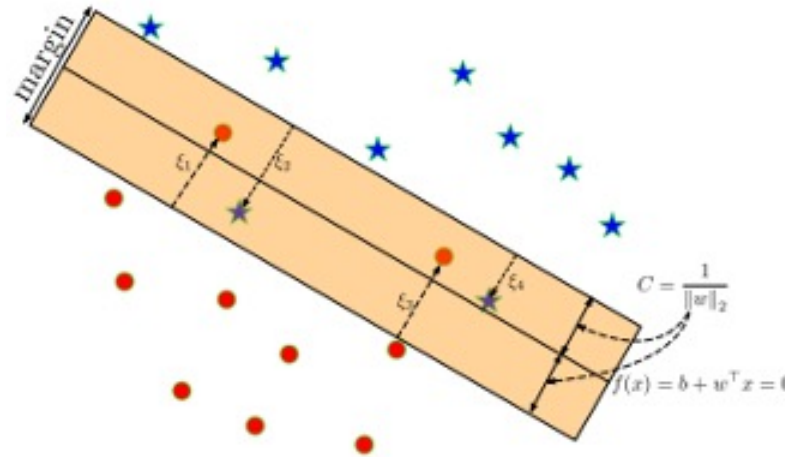
Maximizing Margin for Non-Separable Data Points



Main idea:

- Allow some points to be within the margin or even be misclassified; we represent this with **slack variables** ξ_i . 松弛变量
- But constrain or penalize the total amount of slack.

Maximizing Margin for Non-Separable Data Points



- **Soft margin constraint:**

$$\frac{t^{(i)}(\mathbf{w}^T \mathbf{x}^{(i)} + b)}{\|\mathbf{w}\|_2} \geq C(1 - \xi_i),$$

C < 0 数

for $\xi_i \geq 0$.

- Penalize $\sum_i \xi_i$.

Maximizing Margin for Non-Separable Data Points

Soft-margin SVM objective:

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|_2^2 + \gamma \sum_{i=1}^N \xi_i$$

第 i 个样本的松弛变量

$$\text{s. t. } t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b) \geq 1 - \xi_i \quad i = 1, \dots, N$$

对每个 i 而言

$$\xi_i \geq 0 \quad i = 1, \dots, N$$

- γ is a hyperparameter that trades off the margin with the amount of slack.

- For $\gamma = 0$, we'll get $\mathbf{w} = 0$. (why?) w=0, trust region 恒成立
- As $\gamma \rightarrow \infty$, we get the hard-margin objective.

- Note: it is also possible to constrain $\sum_i \xi_i$ instead of penalizing it.
- ~ $\rightarrow \xi_i \rightarrow 0 \rightarrow$ 回到 hard

From Margin Violation to Hinge Loss

Let's simplify the soft margin constraint by eliminating ξ_i . Recall:

$$t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b) \geq 1 - \xi_i \quad i = 1, \dots, N$$

$$\xi_i \geq 0 \quad i = 1, \dots, N$$

- Rewrite as $\xi_i \geq 1 - t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b)$. $\rightarrow$ 分类讨论
- **Case 1:** $1 - t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b) \leq 0$
 - The smallest non-negative ξ_i that satisfies the constraint is $\xi_i = 0$.
- **Case 2:** $1 - t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b) > 0$
 - The smallest ξ_i that satisfies the constraint is $\xi_i = 1 - t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b)$.
- Hence, $\xi_i = \max\{0, 1 - t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b)\}$.
- Therefore, the slack penalty can be written as

$$\sum_{i=1}^N \xi_i = \sum_{i=1}^N \max\{0, 1 - t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b)\}$$

From Margin Violation to Hinge Loss

If we write $y^{(i)}(\mathbf{w}, b) = \mathbf{w}^\top \mathbf{x} + b$, then the optimization problem can be written as

$$\min_{\mathbf{w}, b, \xi} \sum_{i=1}^N \max\{0, 1 - t^{(i)} y^{(i)}(\mathbf{w}, b)\} + \frac{1}{2\gamma} \|\mathbf{w}\|_2^2$$

- The loss function $\mathcal{L}_H(y, t) = \max\{0, 1 - ty\}$ is called the **hinge** loss. 铰链损失
- The second term is the L_2 -norm of the weights.
- Hence, the soft-margin SVM can be seen as a linear classifier with hinge loss and an L_2 regularizer.

Multiclass SVM Loss

Suppose: 3 training examples, 3 classes.

With some W the scores $f(x, W) = Wx$ are:



cat	3.2	1.3	2.2
car	5.1	4.9	2.5
frog	-1.7	2.0	-3.1

Multiclass SVM loss:

Given an example (x_i, y_i) , where x_i is the image and y_i is the (integer) label.

and using the shorthand for the scores vector: $s = f(x_i, W)$

The SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

$1 - ty$

Multiclass SVM Loss

$wx+b$ 表达的 $S_{\text{正确数}} - S_{\text{错误数}}$

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Losses:	2.9		

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$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

$$\begin{aligned}
 &= \max(0, 5.1 - 3.2 + 1) \\
 &\quad + \max(0, -1.7 - 3.2 + 1) \\
 &= \max(0, 2.9) + \max(0, -3.9) \\
 &= 2.9 + 0 \\
 &= 2.9
 \end{aligned}$$

Multiclass SVM Loss

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The SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

$$\begin{aligned}
 &= \max(0, 1.3 - 4.9 + 1) \\
 &\quad + \max(0, 2.0 - 4.9 + 1) \\
 &= \max(0, -2.6) + \max(0, -1.9) \\
 &= 0 + 0 \\
 &= 0
 \end{aligned}$$

Multiclass SVM Loss

Suppose: 3 training examples, 3 classes.

With some W the scores $f(x, W) = Wx$ are:



cat	3.2	1.3	2.2
car	5.1	4.9	2.5
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Losses:	2.9	0	12.9

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(integer) label.

and using the shorthand for the scores
vector: $s = f(x_i, W)$

The SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

$$\begin{aligned}
 &= \max(0, 2.2 - (-3.1) + 1) \\
 &\quad + \max(0, 2.5 - (-3.1) + 1) \\
 &= \max(0, 6.3) + \max(0, 6.6) \\
 &= 6.3 + 6.6 \\
 &= 12.9
 \end{aligned}$$

Multiclass SVM Loss

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With some W the scores $f(x, W) = Wx$ are:



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The SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

and the full training loss is the mean over
all examples in the training data:

$$L = \frac{1}{N} \sum_{i=1}^N L_i$$

$$L = (2.9 + 0 + 12.9)/3$$

$$= 5.27$$

Multiclass SVM Loss

Suppose: 3 training examples, 3 classes.

With some W the scores $f(x, W) = Wx$ are:



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Multiclass SVM loss:

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(integer) label.

and using the shorthand for the scores
vector: $s = f(x_i, W)$

The SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

Q: what if the sum was instead over all
classes? (including $j=y_i$)

Multiclass SVM Loss

Suppose: 3 training examples, 3 classes.

With some W the scores $f(x, W) = Wx$ are:



cat	3.2	1.3	2.2
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vector: $s = f(x_i, W)$

The SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

Q2: what if we used a mean instead of a
sum here

no influence for train

Loss ↓

Multiclass SVM Loss

Suppose: 3 training examples, 3 classes.

With some W the scores $f(x, W) = Wx$ are:



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vector: $s = f(x_i, W)$

The SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

Q3: what if we used

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)^2$$

Loss change, training not

Multiclass SVM Loss

Suppose: 3 training examples, 3 classes.

With some W the scores $f(x, W) = Wx$ are:



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vector: $s = f(x_i, W)$

The SVM loss has the form:

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Q4: what is the min/max possible loss

Multiclass SVM Loss

Suppose: 3 training examples, 3 classes.

With some W the scores $f(x, W) = Wx$ are:



cat	3.2	1.3	2.2
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Losses:	2.9	0	12.9

Multiclass SVM loss:

Given an example (x_i, y_i) , where x_i is the image and y_i is the (integer) label.

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The SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

Q5: usually at initialization W are small numbers so all $s \approx 0$. What is the loss?

Softmax

Softmax Classifier (Multinomial Logistic Regression)



cat	3.2
car	5.1
frog	-1.7

Softmax

Softmax Classifier (Multinomial Logistic Regression)



scores = unnormalized log probabilities of the classes.

$$s = f(x_i; W)$$

cat	3.2
car	5.1
frog	-1.7

Softmax

Softmax Classifier (Multinomial Logistic Regression)



scores = unnormalized log probabilities of the classes.

$$P(Y = k|X = \mathbf{x}_i) = \frac{e^{s_k}}{\sum_j e^{s_j}} \text{ where } \boxed{s = f(\mathbf{x}_i; W)}$$

cat	3.2
car	5.1
frog	-1.7

Softmax

Softmax Classifier (Multinomial Logistic Regression)



scores = unnormalized log probabilities of the classes.

$$P(Y = k|X = \mathbf{x}_i) = \frac{e^{s_k}}{\sum_j e^{s_j}} \text{ where } s = f(\mathbf{x}_i; W)$$

Softmax function

cat	3.2
car	5.1
frog	-1.7

Softmax

Softmax Classifier (Multinomial Logistic Regression)



scores = unnormalized log probabilities of the classes.

$$P(Y = k|X = \mathbf{x}_i) = \frac{e^{s_k}}{\sum_j e^{s_j}} \text{ where } s = f(\mathbf{x}_i; W)$$

cat	3.2
car	5.1
frog	-1.7

Want to maximize the log likelihood, or (for a loss function) to minimize the negative log likelihood of the correct class:

$$L_i = -\log P(Y = \mathbf{y}_i|X = \mathbf{x}_i)$$

Softmax

Softmax Classifier (Multinomial Logistic Regression)



cat

3.2

car

5.1

frog

-1.7

unnormalized log probabilities

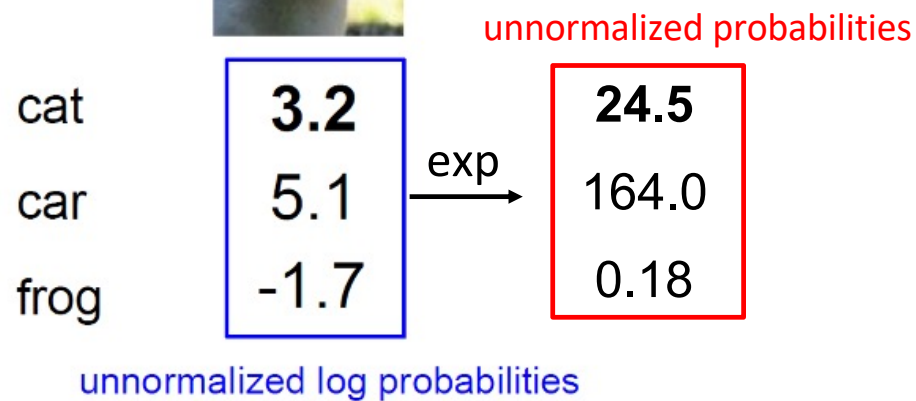
$$L_i = -\log\left(\frac{e^{s_{y_i}}}{\sum_j e^{s_j}}\right)$$

Softmax

Softmax Classifier (Multinomial Logistic Regression)

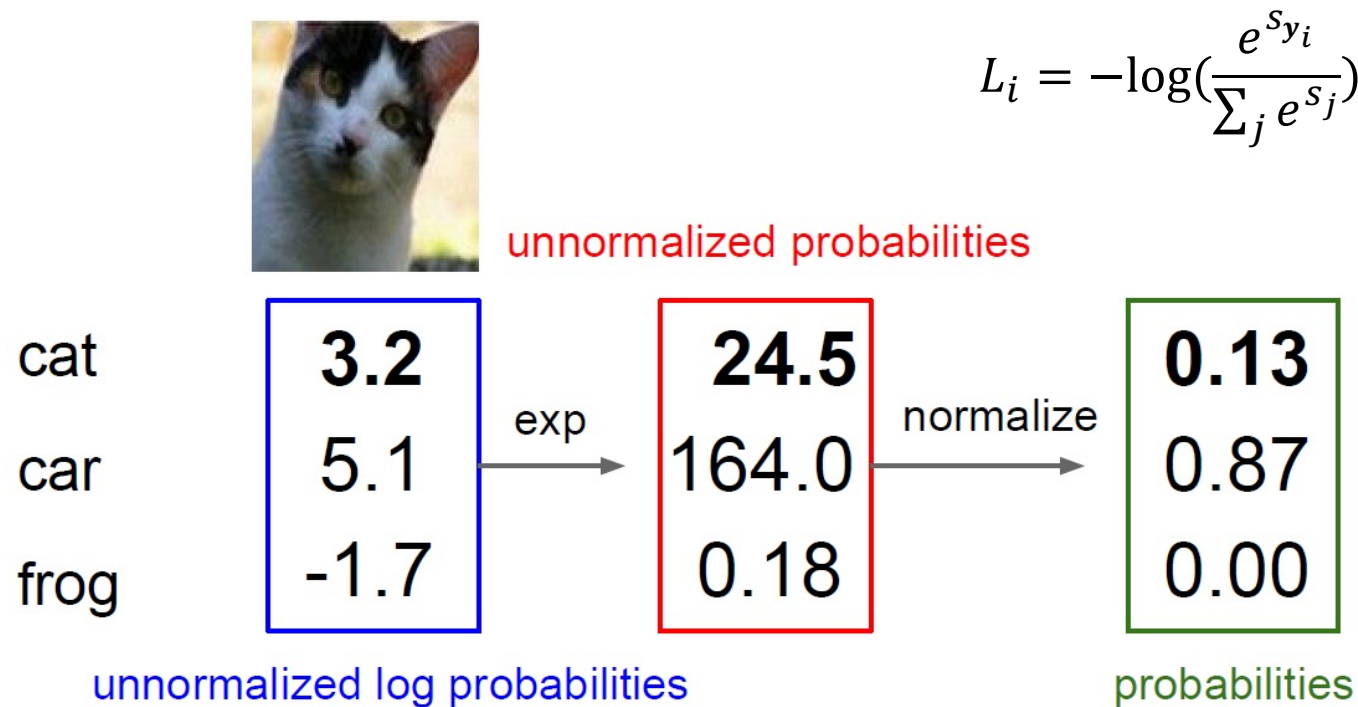


$$L_i = -\log\left(\frac{e^{s_{y_i}}}{\sum_j e^{s_j}}\right)$$



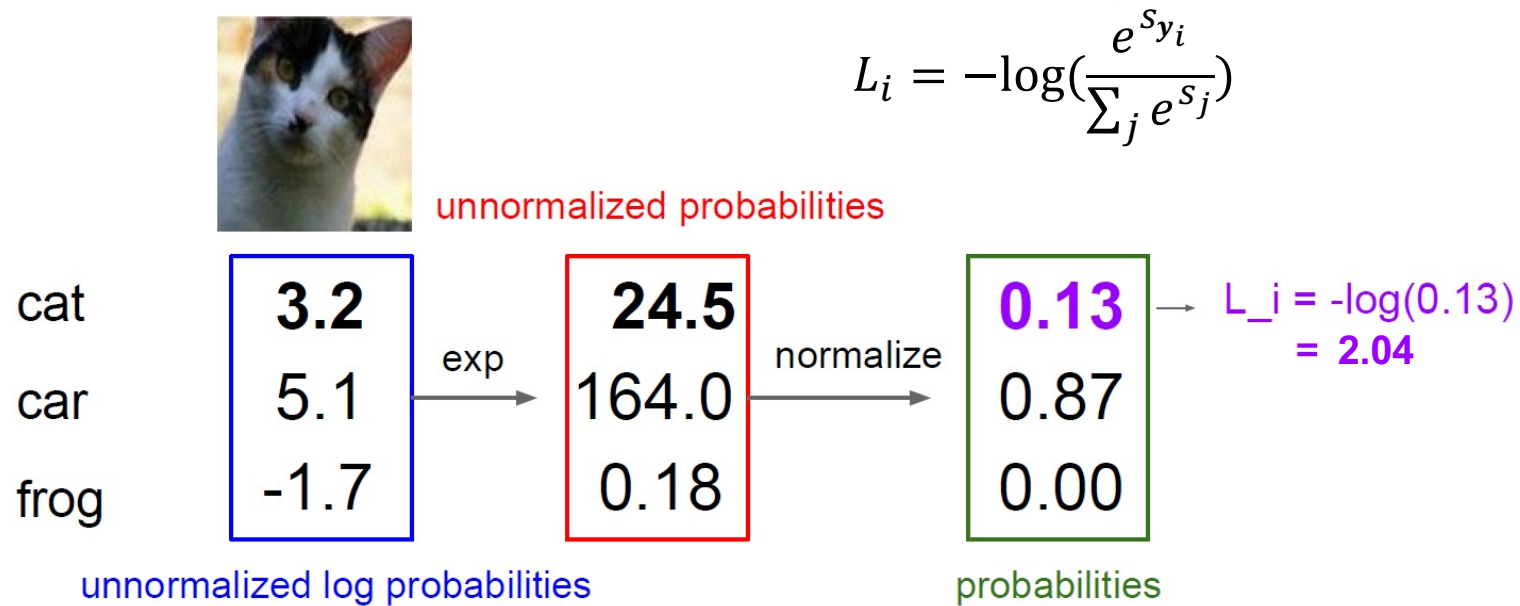
Softmax

Softmax Classifier (Multinomial Logistic Regression)



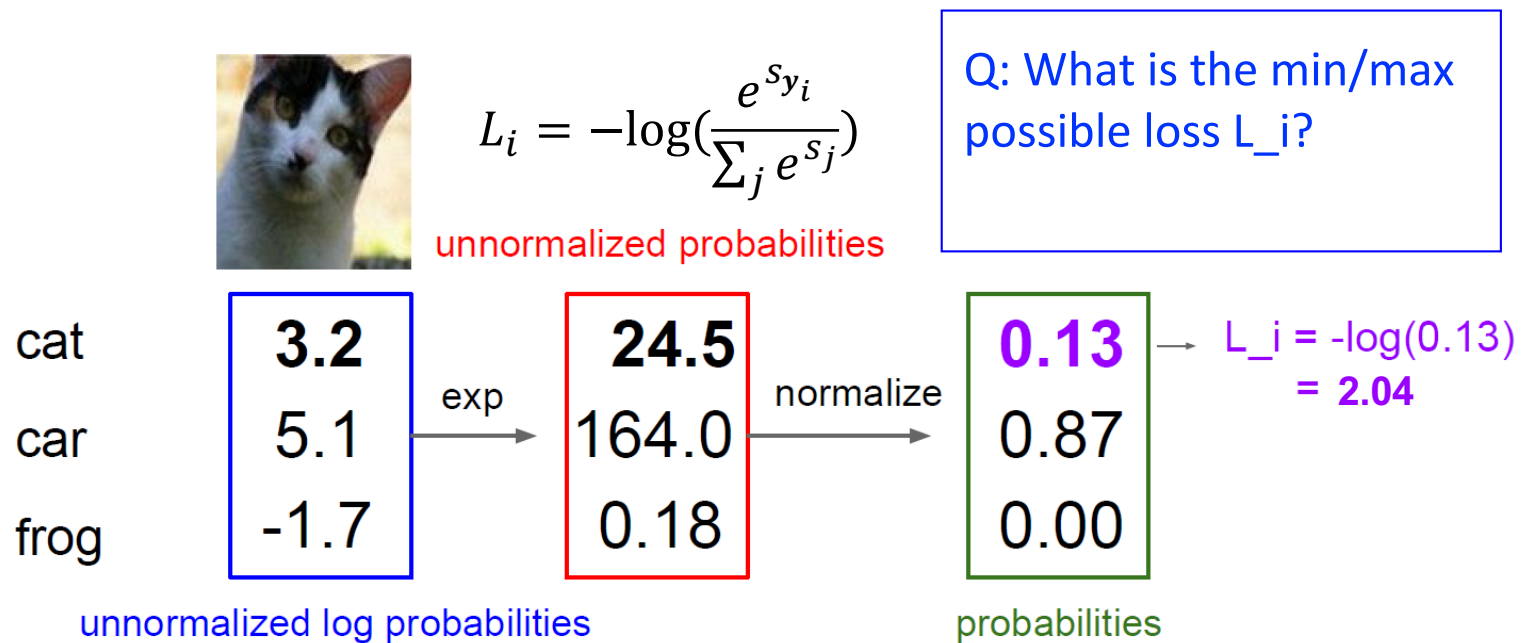
Softmax

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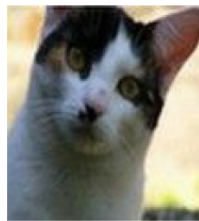
Softmax

Softmax Classifier (Multinomial Logistic Regression)



Softmax

Softmax Classifier (Multinomial Logistic Regression)



$$L_i = -\log\left(\frac{e^{s_{y_i}}}{\sum_j e^{s_j}}\right)$$

unnormalized probabilities

Q5: usually at initialization W are small numbers, so all $s \approx 0$. What is the loss?

cat
car
frog

3.2
5.1
-1.7

exp

24.5
164.0
0.18

normalize

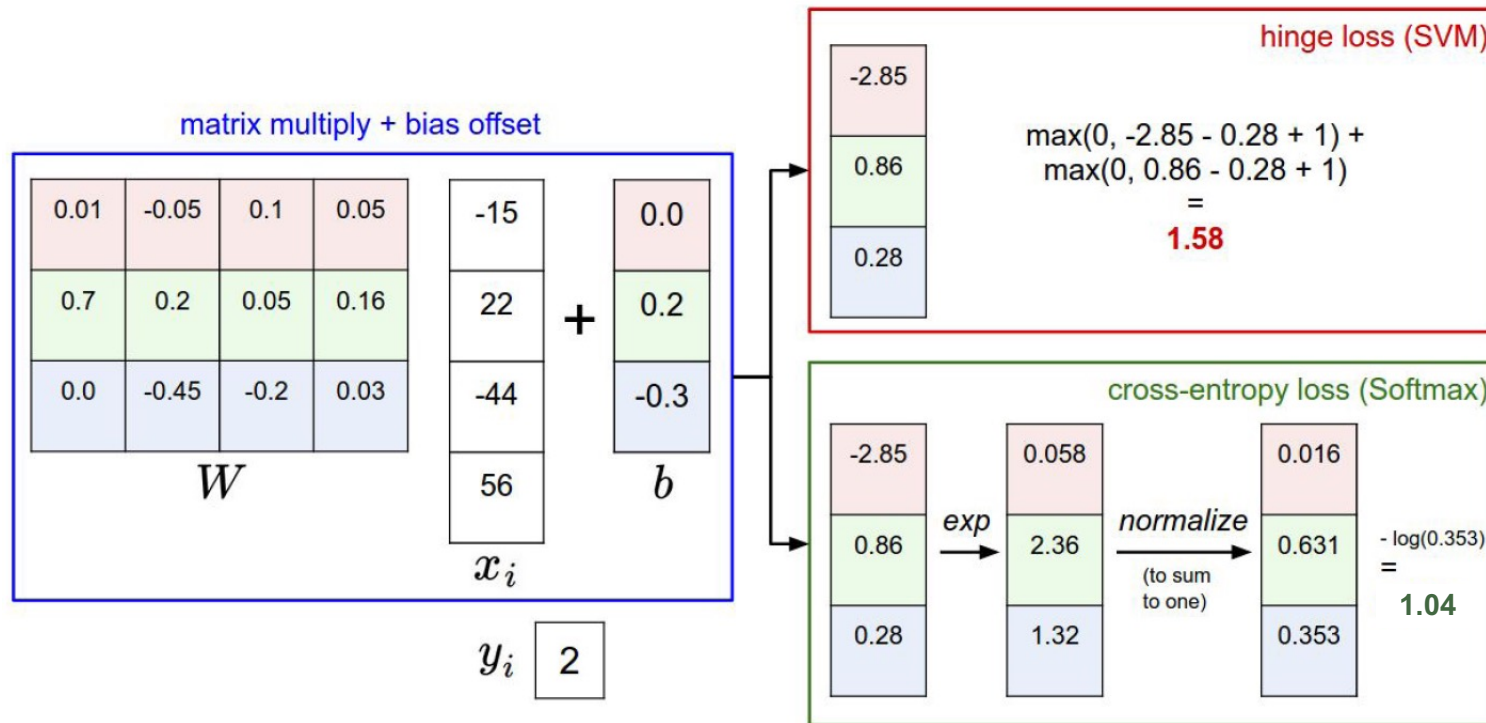
0.13
0.87
0.00

$$L_i = -\log(0.13) = 2.04$$

unnormalized log probabilities

probabilities

SVM & Softmax



SVM & Softmax

Softmax vs. SVM

$$L_i = -\log\left(\frac{e^{s_{y_i}}}{\sum_j e^{s_j}}\right)$$

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

SVM & Softmax

Softmax vs. SVM

$$L_i = -\log\left(\frac{e^{s_{y_i}}}{\sum_j e^{s_j}}\right)$$

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

assume scores:

[10, -2, 3]

[10, 9, 9]

[10, -100, -100]

and $y_i = 0$

Q: suppose I take a datapoint and I jiggle a bit (changing its score slightly). What happens to the loss in both cases?